

Question 11

..... finds a path that starts and ends with the same vertex and passes through other vertices once

☐ I. Digraph

☐ II. BFS

☒ III. Topological Sorting

☐ IV. DFS



Question 4

One of the best-known examples of NP-hard problems is

- ☐ I. Assignment problem
- ☐ II. Topological sorting
- ☒ III. Knapsack problem
- ☐ IV. None

→ ⚠ Moving to another question will save this response.

Question 14

Finding articulation points is one of applications.

- ☐ I. Closest Pair problem
- ☐ II. Convex Hull problem
- ☐ III. BFS
- ☐ IV. DFS

Question 15

Fake-Coin problem is

- ☐ I. A linearithmic time algorithm
- ☐ II. A factorial time algorithm
- ☒ III. A logarithmic time algorithm
- ☐ IV. An exponential time algorithm

Question 13

The total number of permutations needed in the traveling salesman problem is

☐ I. $(n-1)!$

☒ II. $0.5(n-1)!$

☐ III. $0.5n!$

☐ IV. $n!$

Question 3

0.5 points

If S is a set that contains 4 elements $\{a, b, c, d\}$, what is the bit string that represents the subset $\{a, c, d\}$?

☐ I. 0101

☐ II. 1110

☒ III. 1011

☐ IV. 0111

⏪ ⚠ Moving to another question will save this response.

0.75 points

Save Answer

Question 1

Apply Strassen's algorithm to compute the product of the two matrices, $A * B$, and fill in the blanks.

$$\begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 9 & -3 \\ 6 & 1 \end{bmatrix}$$

- The value of **m1** is equal to
- The value of **m3** is equal to
- The value of **m5** is equal to

20

4

3

⏪ ⚠ Moving to another question will save this response.

Question 7

0.25 points

In the traversal, the root is visited after visiting its left subtree but before visiting the right subtree.

☐ I. Post-order

☒ II. In-order

☐ III. Pre-order

☐ IV. Level-order

Question 8

..... algorithm has a $n \log n$ worst-case time efficiency.

☒ I. Mergesort

☐ II. Bubble sort

☐ III. Quicksort

☐ IV. Insertion sort

Question 2

Apply the Master theorem to find the order of growth for the following recurrence:

$$T(n) = 4T(n/2) + n^3$$

☐ I. $n^3 \log n$

☐ II. n^2

☒ III. n^3

☐ IV. $n \log_2^3 n$

Question 12

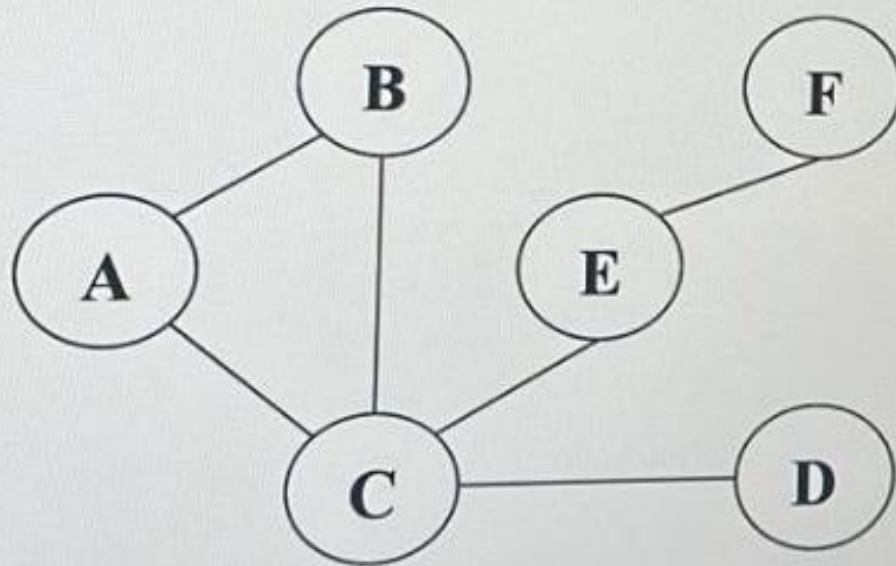
Quickselect algorithm used to

- ☐ I. Find the kth smallest element in a list
- ☐ II. Compute median
- ☐ III. Find the kth largest element in a list
- ☐ IV. All of the Above

Question 9

What is the content of the queue after traversing the following graph using the BFS algorithm?

[Note: You must start from vertex (A) and break the ties alphabetically.]



- ☐ I. A, C, D, B, E, F
- ☐ II. A, B, C, E, F, D
- ☐ III. A, C, E, F, B, D
- ☒ IV. A, B, C, D, E, F

Question 10

We can enhance Quicksort algorithm by

- ☐ I. Three-way partition
- ☐ II. Median-of-three
- ☐ III. Randomize Quicksort
- ☐ IV. All of the above

Question 6

Using the Lexicographic algorithm, the next permutation of (1 2 4 3 5) is

☒ I. 1 2 4 5 3

☐ II. 1 3 2 4 5

☐ III. 1 4 2 3 5

☐ IV. 1 5 2 3 5

Question 5

Binary Search is a type of algorithm.

- ☐ I. Decrease-and-conquer (variable size decrease)
- ☒ II. Decrease-and-conquer (by a constant factor)
- ☐ III. Divide-and-conquer
- ☐ IV. Decrease-and-conquer (by a constant)